

SQL Server — Class 2 Homework

Joins & Views

BikeStores Sample Database

Instructions

Write a SQL Server query to answer each question below using the BikeStores sample database.

Tables: *production.products* | *production.brands* | *production.categories* | *sales.customers* | *sales.orders* | *sales.order_items* | *sales.staffs* | *sales.stores* | *sales.stocks*

Topic Legend:

INNER JOIN

LEFT JOIN

RIGHT JOIN

SELF JOIN

VIEWS

BONUS

Q1

Show all product names along with their brand name. Sort by brand name, then by product name alphabetically.

INNER JOIN

Your answer:

SQL Query1.sql...ujitsu (52))*

```

2
3
4 SELECT
5     p.product_name,
6     b.brand_name
7 FROM
8     production.products p
9 JOIN
10    production.brands b
11    ON p.brand_id = b.brand_id
12 ORDER BY
13     b.brand_name ASC,
14     p.product_name ASC;

```

100 %
No issues found

Results
Messages

	product_name	brand_name
1	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	Electra
2	Electra Amsterdam Fashion 7i Ladies' - 2017	Electra
3	Electra Amsterdam Original 3i - 2015/2017	Electra
4	Electra Amsterdam Original 3i Ladies' - 2017	Electra
5	Electra Amsterdam Royal 8i - 2017/2018	Electra
6	Electra Amsterdam Royal 8i Ladies - 2018	Electra
7	Electra Cruiser 1 (24-Inch) - 2016	Electra

Query execut...
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Q2

List all products with their category name and list price. Sort by category name, then by price from cheapest to most expensive.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (52))*

```
1
2
3 SELECT
4     p.product_name,
5     c.category_name,
6     p.list_price
7 FROM
8     production.products p
9 JOIN
10    production.categories c
11    ON p.category_id = c.category_id
12 ORDER BY
13     c.category_name ASC,
14     p.list_price ASC;
```

100 % No issues found

Results Messages

	product_name	category_name	list_price
1	Strider Classic 12 Balance Bike - 2018	Children Bicycles	89.99
2	Sun Bicycles Lil Kitt'n - 2017	Children Bicycles	109.99
3	Trek Boy's Kickster - 2015/2017	Children Bicycles	149.99
4	Trek Girl's Kickster - 2017	Children Bicycles	149.99
5	Trek Kickster - 2018	Children Bicycles	159.99
6	Trek Precaliber 12 Boys - 2017	Children Bicycles	189.99
7	Trek Precaliber 12 Girls - 2017	Children Bicycles	189.99
8	Trek Precaliber 12 Boy's - 2018	Children Bicycles	199.99

Query executed successfully.

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Q3

Show all orders with the customer's full name and order date. Sort by order date from newest to oldest.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```
1
2  --Show all orders with the customer's full name and order date.
3  --Sort by order date from newest to oldest.
4
5
6  SELECT
7      o.order_id,
8      c.first_name + ' ' + c.last_name AS full_name,
9      o.order_date
10
11  FROM
12      sales.orders o
13  JOIN
14      sales.customers c
15      ON o.customer_id = c.customer_id
16  ORDER BY
17      o.order_date DESC;
```

100 %

✓ No issues found

Results Messages

	order_id	full_name	order_date
1	1615	Sarita Parks	2018-12-28
2	1614	Dorthey Jackson	2018-11-28
3	1613	Debra Burks	2018-11-18
4	1612	Tameka Fisher	2018-10-21
5	1611	Lyndsey Bean	2018-09-06
6	1610	Linnie Branch	2018-08-25
7	1609	Pamelia Newman	2018-08-23
8	1608	Satamina Gamer	2018-07-12

✓ Query executed successfully.

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Q4

Display each order item with the product name, quantity, unit price, and a computed column called "Line Total" (quantity × list_price). Sort by order ID.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```
1
2  --Display each order item with the product name,
3  -- quantity, unit price, and a computed column called "Line Total" (quantity * list_price).
4  --Sort by order ID.
5
6  SELECT
7      oi.order_id,
8      p.product_name,
9      oi.quantity,
10     oi.list_price AS unit_price,
11     (oi.quantity * oi.list_price) AS [Line Total]
12 FROM
13     sales.order_items oi
14 JOIN
15     production.products p
16     ON oi.product_id = p.product_id
17 ORDER BY
18     oi.order_id ASC;
19
```

100 %

✓ No issues found

Ln:

Results Messages

	order_id	product_name	quantity	unit_price	Line Total
1	1	Electra Townie Original 7D EQ - Women's - 2016	1	599.99	599.99
2	1	Trek Remedy 29 Carbon Frameset - 2016	2	1799.99	3599.98
3	1	Surly Straggler - 2016	2	1549.00	3098.00
4	1	Electra Townie Original 7D EQ - 2016	2	599.99	1199.98
5	1	Trek Fuel EX 8 29 - 2016	1	2899.99	2899.99
6	2	Electra Townie Original 7D EQ - Women's - 2016	1	599.99	599.99
7	2	Electra Townie Original 7D EQ - 2016	2	599.99	1199.98

✓ Query executed successfully.

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BikeStores

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Q5

Show each order along with the store name where it was placed and the order date. Sort by store name.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```
1
2  --Show each order along with the store name
3  --where it was placed and the order date.Sort by store name.
4
5  SELECT
6      o.order_id,
7      s.store_name,
8      o.order_date
9  FROM
10     sales.orders o
11  JOIN
12     sales.stores s
13     ON o.store_id = s.store_id
14  ORDER BY
15     s.store_name ASC;
16
```

100 %

✓ No issues found

Results Messages

	order_id	store_name	order_date
1	2	Baldwin Bikes	2016-01-01
2	3	Baldwin Bikes	2016-01-02
3	5	Baldwin Bikes	2016-01-03
4	6	Baldwin Bikes	2016-01-04
5	7	Baldwin Bikes	2016-01-04
6	8	Baldwin Bikes	2016-01-04
7	10	Baldwin Bikes	2016-01-05

✓ Query executed successfully.

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Q6

Show each order with: order ID, customer full name, store name, and the staff member's full name who handled it.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

2

--Show each order with: order ID, customer full name,

3

--store name, and the staff member's full name who handled it.

4

5

SELECT

6

o.order_id,

7

c.first_name + ' ' + c.last_name AS customer_name,

8

s.store_name,

9

st.first_name + ' ' + st.last_name AS staff_name

10

FROM

11

sales.orders o

12

JOIN

13

sales.customers c

14

ON o.customer_id = c.customer_id

15

JOIN

16

sales.stores s

17

ON o.store_id = s.store_id

18

JOIN

19

sales.staffs st

20

ON o.staff_id = st.staff_id;

21

100 %

No issues found

Results

Messages

	order_id	customer_name	store_name	staff_name
1	1	Johnathan Velazquez	Santa Cruz Bikes	Mireya Copeland
2	2	Jaqueline Cummings	Baldwin Bikes	Marcelene Boyer
3	3	Joshua Robertson	Baldwin Bikes	Venita Daniel
4	4	Nova Hess	Santa Cruz Bikes	Genna Serrano
5	5	Arla Ellis	Baldwin Bikes	Marcelene Boyer
6	6	Sharyn Hopkins	Baldwin Bikes	Marcelene Boyer
7	7	Laureen Paul	Baldwin Bikes	Marcelene Boyer

Query executed successfully.

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Q7

List all products from the brand "Trek" along with their category name and price. Sort by price descending. (Use JOIN — do NOT filter by brand_id directly.)

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```
2  --List all products from the brand "Trek" along with their category name and price.
3  --Sort by price descending.(Use JOIN - do NOT filter by brand_id directly.)
4
5  SELECT
6      p.product_name,
7      c.category_name,
8      p.list_price
9  FROM
10     production.products p
11  JOIN
12     production.brands b
13     ON p.brand_id = b.brand_id
14  JOIN
15     production.categories c
16     ON p.category_id = c.category_id
17  WHERE
18     b.brand_name = 'Trek'
19  ORDER BY
20     p.list_price DESC;|
21
```

100 %

✓ No issues found

Results Messages

	product_name	category_name	list_price
1	Trek Domane SLR 9 Disc - 2018	Road Bikes	11999.99
2	Trek Domane SLR 8 Disc - 2018	Road Bikes	7499.99
3	Trek Domane SL Frameset - 2018	Road Bikes	6499.99
4	Trek Domane SL Frameset Women's - 2018	Road Bikes	6499.99
5	Trek Silque SLR 8 Women's - 2017	Road Bikes	6499.99
6	Trek Emonda SLR 8 - 2018	Road Bikes	6499.99
7	Trek Silque SLR 7 Women's - 2017	Road Bikes	5999.99

✓ Query executed successfully.

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Q8

Find all customers from the state of "NY" who have placed at least one order. Show customer full name, city, and order date. (Use JOIN — do not use a subquery.)

INNER JOIN

Your answer:

```
SQLQuery1.sql...ujitsu (51))* ✕ X
2  --Find all customers from the state of "NY" who have placed at least one order.
3  --Show customer full name,city, and order date. (Use JOIN – do not use a subquery)
4
5  SELECT
6      c.first_name + ' ' + c.last_name AS customer_name,
7      c.city,
8      o.order_date
9  FROM
10     sales.customers c
11  JOIN
12     sales.orders o
13     ON c.customer_id = o.customer_id
14  WHERE
15     c.state = 'NY'
16  ORDER BY
17     o.order_date ASC;
18
19
```

100 % ✓ No issues found

Results Messages

	customer_name	city	order_date
1	Jaqueline Cummings	Huntington Station	2016-01-01
2	Joshua Robertson	Patchogue	2016-01-02
3	Arla Ellis	Utica	2016-01-03
4	Leslie Higgins	Saratoga Springs	2016-01-04
5	Sharyn Hopkins	Baldwinsville	2016-01-04
6	Laureen Paul	Bellmore	2016-01-04
7	Alane Munoz	Yonkers	2016-01-05

✓ Query executed successfully.

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Q9

Show all completed orders (order_status = 4) from the store "Rowlett Bikes". Display order ID, customer full name, and order date.

INNER JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```
2  --Show all completed orders (order_status = 4) from the store "Rowlett Bikes".
3  --Display order ID, customer full name, and order date.
4
5  SELECT
6      o.order_id,
7      c.first_name + ' ' + c.last_name AS customer_name,
8      o.order_date
9  FROM
10     sales.orders o
11  JOIN
12     sales.customers c
13     ON o.customer_id = c.customer_id
14  JOIN
15     sales.stores s
16     ON o.store_id = s.store_id
17  WHERE
18     o.order_status = 4
19     AND s.store_name = 'Rowlett Bikes'
20  ORDER BY
21     o.order_date ASC;
```

100 %

✓ No issues found

Results Messages

	order_id	customer_name	order_date
1	31	Edgar Quinn	2016-01-20
2	50	Silas Tate	2016-01-31
3	67	Lazaro Moran	2016-02-09
4	89	Calandra Stanton	2016-02-21
5	99	Rikki Morrow	2016-02-29
6	101	Katheleen Marks	2016-03-01
7	106	Valery Saunders	2016-03-04

✓ Query executed successfully.

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Q10

List ALL customers and any orders they have placed. Include customers who have never placed an order (show NULL for order columns). Sort by customer ID.

LEFT JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```

2  --List ALL customers and any orders they have placed.
3  --Include customers who have never placed an order (show NULL for order columns).
4  --Sort by customer ID.
5
6  SELECT
7      c.customer_id,
8      c.first_name + ' ' + c.last_name AS customer_name,
9      o.order_id,
10     o.order_date
11 FROM
12     sales.customers c
13 LEFT JOIN
14     sales.orders o
15     ON c.customer_id = o.customer_id
16 ORDER BY
17     c.customer_id ASC;
18
19
20

```

100 % No issues found

Results Messages

	customer_id	customer_name	order_id	order_date
1	1	Debra Burks	599	2016-12-09
2	1	Debra Burks	1555	2018-04-18
3	1	Debra Burks	1613	2018-11-18
4	2	Kasha Todd	1509	2018-04-09
5	2	Kasha Todd	692	2017-02-05
6	2	Kasha Todd	1084	2017-08-21
7	3	Tameka Fisher	1496	2018-04-06

Query executed successfully.

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Q11

Find all customers who have NEVER placed an order. Show their full name and email.

LEFT JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

2

--Find all customers who have NEVER placed an order.

3

--Show their full name and email.

4

5

SELECT

6

c.first_name + ' ' + c.last_name AS customer_name,

7

c.email

8

FROM

9

sales.customers c

10

LEFT JOIN

11

sales.orders o

12

ON c.customer_id = o.customer_id

13

WHERE

14

o.order_id IS NULL;

15

16

100 %

No issues found

Results

Messages

customer_name	email
---------------	-------

Query executed successfully.

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Q12

List all products and their stock quantity at every store. Include products that have NO stock record at all. Show product name, store ID, and quantity.

LEFT JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```

2  --List all products and their stock quantity at every store.
3  --Include products that have NO stock record at all.
4  --Show product name, store ID, and quantity.
5
6  SELECT
7      p.product_name,
8      s.store_id,
9      s.quantity
10 FROM
11     production.products p
12 LEFT JOIN
13     production.stocks s
14     ON p.product_id = s.product_id
15 ORDER BY
16     p.product_name;
17
18

```

100 %

✓ No issues found

Results

Messages

	product_name	store_id	quantity
1	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	1	17
2	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	2	5
3	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	3	25
4	Electra Amsterdam Fashion 7i Ladies' - 2017	1	25
5	Electra Amsterdam Fashion 7i Ladies' - 2017	2	19
6	Electra Amsterdam Fashion 7i Ladies' - 2017	3	18
7	Electra Amsterdam Original 3i - 2015/2017	1	3

✓ Query executed successfully.

DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-2

Q13

Find all products that have NEVER been ordered (no record in order_items). Show product name and list price.

LEFT JOIN

Your answer:

The screenshot shows a SQL Developer window titled "SQLQuery1.sql...ujitsu (51))*". The query is as follows:

```
1
2  --Find all products that have NEVER been ordered (no record in order_items).
3  --Show product name and list price.
4
5  SELECT
6      p.product_name,
7      p.list_price
8  FROM
9      production.products p
10 LEFT JOIN
11     sales.order_items oi
12     ON p.product_id = oi.product_id
13 WHERE
14     oi.product_id IS NULL
15 ORDER BY
16     p.product_name;
17
18
```

Below the query editor, the status bar shows "100 %", a green checkmark, and "No issues found". The "Results" tab is active, displaying a table with 7 rows and 2 columns: "product_name" and "list_price".

	product_name	list_price
1	Electra Savannah 1 (20-inch) - Girl's - 2018	319.99
2	Electra Sweet Ride 1 (20-inch) - Girl's - 2018	319.99
3	Electra Townie Go! 8i Ladies' - 2018	2599.99
4	Surly Krampus Frameset - 2018	2499.99
5	Trek 820 - 2016	379.99
6	Trek Checkpoint ALR 4 Women's - 2019	1699.99
7	Trek Checkpoint ALR 5 - 2019	1999.99

At the bottom, a yellow status bar indicates "Query executed successfully." followed by a lock icon and the text "DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fu".

Q14

List each staff member along with the full name of their manager. Staff with no manager (top-level) should still appear — show NULL for manager name.

SELF JOIN

Your answer:

SQLQuery1.sql...ujitsu (51))*

```

1
2  --List each staff member along with the full name of their manager.
3  --Staff with no manager (top-level) should still appear - show NULL for manager name.
4
5  SELECT
6      s.first_name + ' ' + s.last_name AS staff_name,
7      m.first_name + ' ' + m.last_name AS manager_name
8  FROM
9      sales.staffs s
10 LEFT JOIN
11     sales.staffs m
12     ON s.manager_id = m.staff_id
13 ORDER BY
14     staff_name;
15
16

```

100 % No issues found

Results Messages

	staff_name	manager_name
1	Bernardine Houston	Venita Daniel
2	Fabiola Jackson	NULL
3	Genna Serrano	Mireya Copeland
4	Jannette David	Fabiola Jackson
5	Kali Vargas	Fabiola Jackson
6	Layla Terrell	Venita Daniel
7	Marcelene Boyer	Jannette David

Query executed successfully.

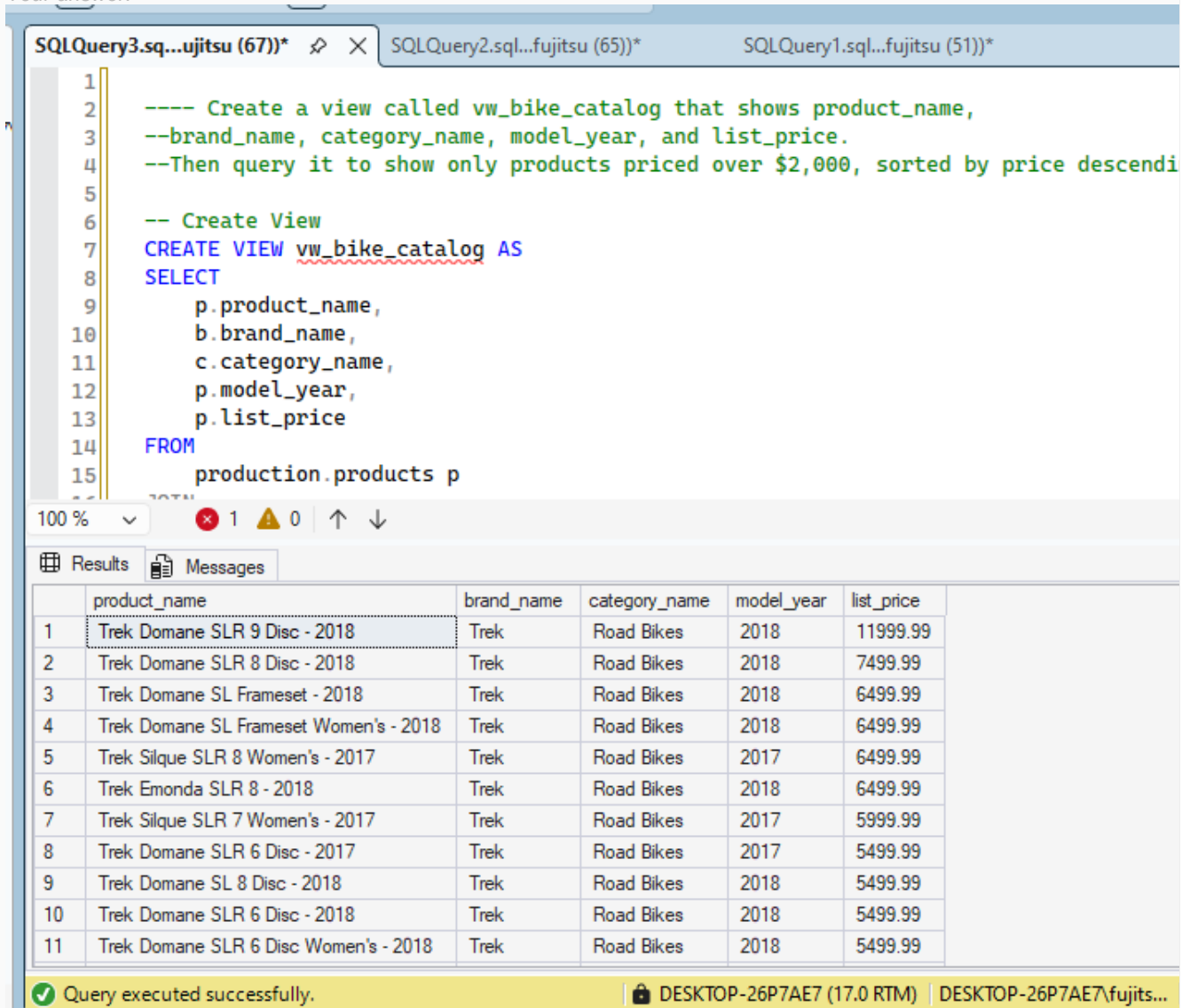
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Q15

Create a view called vw_bike_catalog that shows product_name, brand_name, category_name, model_year, and list_price. Then query it to show only products priced over \$2,000, sorted by price descending.

VIEWS

Your answer:



```
1
2  ---- Create a view called vw_bike_catalog that shows product_name,
3  --brand_name, category_name, model_year, and list_price.
4  --Then query it to show only products priced over $2,000, sorted by price descendi
5
6  -- Create View
7  CREATE VIEW vw_bike_catalog AS
8  SELECT
9      p.product_name,
10     b.brand_name,
11     c.category_name,
12     p.model_year,
13     p.list_price
14 FROM
15     production.products p
```

	product_name	brand_name	category_name	model_year	list_price
1	Trek Domane SLR 9 Disc - 2018	Trek	Road Bikes	2018	11999.99
2	Trek Domane SLR 8 Disc - 2018	Trek	Road Bikes	2018	7499.99
3	Trek Domane SL Frameset - 2018	Trek	Road Bikes	2018	6499.99
4	Trek Domane SL Frameset Women's - 2018	Trek	Road Bikes	2018	6499.99
5	Trek Silque SLR 8 Women's - 2017	Trek	Road Bikes	2017	6499.99
6	Trek Emonda SLR 8 - 2018	Trek	Road Bikes	2018	6499.99
7	Trek Silque SLR 7 Women's - 2017	Trek	Road Bikes	2017	5999.99
8	Trek Domane SLR 6 Disc - 2017	Trek	Road Bikes	2017	5499.99
9	Trek Domane SL 8 Disc - 2018	Trek	Road Bikes	2018	5499.99
10	Trek Domane SLR 6 Disc - 2018	Trek	Road Bikes	2018	5499.99
11	Trek Domane SLR 6 Disc Women's - 2018	Trek	Road Bikes	2018	5499.99

Query executed successfully. | DESKTOP-26P7AE7 (17.0 RTM) | DESKTOP-26P7AE7\fujits...

Q16

BONUS: Create a view called vw_customer_orders showing: customer full name, order_id, order_date, store_name, and order_status. Then query it to show only orders where the customer city is "New York", sorted by order_date.

VIEWS

Your answer:

SQLQuery2.sql...ujitsu (65))* SQLQuery1.sql...fujitsu (51))*

```
1
2 --BONUS: Create a view called vw_customer_orders showing: customer full name,
3 --order_id, order_date, store_name, and order_status. Then query it to
4 --show only orders where the customer city is "New York", sorted by order_date.
5
6 -- Create View
7 CREATE VIEW vw_customer_orders AS
8 SELECT
9     c.first_name + ' ' + c.last_name AS customer_name,
10    c.city,
11    o.order_id,
12    o.order_date,
13    s.store_name,
14    o.order_status
15 FROM
16     sales.customers c
```

100 % No issues found

Results Messages

	customer_name	order_id	order_date	store_name	order_status
1	Sharie Alvarez	120	2016-03-14	Baldwin Bikes	4
2	Phylis Adkins	182	2016-04-18	Baldwin Bikes	4
3	Octavia Case	274	2016-06-17	Baldwin Bikes	4
4	Emmitt Sanchez	352	2016-08-03	Baldwin Bikes	4
5	Guillermo Hart	514	2016-10-19	Baldwin Bikes	4
6	Genoveva Tyler	572	2016-11-24	Baldwin Bikes	4
7	Emmitt Sanchez	1020	2017-07-23	Baldwin Bikes	3
8	Shenna Benton	1351	2018-01-16	Baldwin Bikes	4
9	Emmitt Sanchez	1510	2018-04-09	Baldwin Bikes	2

Query executed successfully.

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Object Explorer

connect

DESKTOP-26P7AE7 (SQL Server 17.0.1000.7 - DESKTOP-26P7A)

- Databases
 - System Databases
 - Database Snapshots
 - BikeStores
 - Database Diagrams
 - Tables
 - Views
 - System Views
 - dbo.vw_bike_catalog
 - dbo.vw_customer_orders
 - Dropped Ledger Views
 - External Resources
 - Synonyms
 - Programmability
 - Query Store
 - Service Broker
 - Storage
 - Security
 - BloodGroup
 - ColorScheme
 - technology
 - testdb
- Security
 - Server Objects
 - Replication
 - Always On High Availability
 - Management
 - Integration Services Catalogs

SQLQuery2.sql...ujitsu (65))*

```
31 | order_status
32 | FROM
33 | vw_customer_orders
34 | WHERE
35 | city = 'New York'
36 | ORDER BY
37 | order_date;
38 |
39 |
40 | SELECT *
41 | FROM [dbo].[vw_customer_orders];
```

100 % No issues found

Results Messages

	customer_name	city	order_id	order_date	store_name	order_status
1	Johnathan Velazquez	Pleasanton	1	2016-01-01	Santa Cruz Bikes	4
2	Jaqueline Cummings	Huntington Station	2	2016-01-01	Baldwin Bikes	4
3	Joshua Robertson	Patchogue	3	2016-01-02	Baldwin Bikes	4
4	Nova Hess	Duarte	4	2016-01-03	Santa Cruz Bikes	4
5	Arla Ellis	Utica	5	2016-01-03	Baldwin Bikes	4
6	Sharyn Hopkins	Baldwinsville	6	2016-01-04	Baldwin Bikes	4
7	Lauren Paul	Bellmore	7	2016-01-04	Baldwin Bikes	4
8	Leslie Higgins	Saratoga Springs	8	2016-01-04	Baldwin Bikes	4
9	Neil McCall	San Carlos	9	2016-01-05	Santa Cruz Bikes	4
10	Alane Munoz	Yonkers	10	2016-01-05	Baldwin Bikes	4

Query executed successfully.

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